Bishal Dhungana

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SUMMARY

Geospatial Data Scientist working across GIS, applied AI/ML, and cloud data platform engineering, with a focus on disaster resilience and critical infrastructure. I combine field survey and remote sensing origins with production software engineering: building AI-driven inspection systems, geospatial data pipelines, and cloud infrastructure for a major electric utility, while maintaining a parallel practice of published open-source research and tools. Comfortable moving fluidly between data science, data engineering, GIS development, and geomatics depending on what a problem needs.

SKILLS

Machine Learning & Computer Vision: PyTorch, TensorFlow/Keras, Ultralytics, Roboflow, YOLO, RT-DETR, Detectron2, Faster/Mask R-CNN, EfficientDet, DETR, MMDetection, ResNet, XGBoost, OpenCV, ONNX

Generative AI & Agentic Systems: LangChain, LangGraph, RAG, Model Context Protocol, FastMCP, Vector Databases, Ollama, Chroma, Hugging Face Transformers

Data Engineering: Python, SQL, ETL/ELT Pipelines, PySpark, DuckDB, Databricks, Data Lakes, Parquet

Geospatial: ArcGIS Pro / Enterprise / Online, Utility Network, PostGIS, DuckDB Spatial, GDAL, PDAL, GeoPandas, Shapely, PyProj, LiDAR / Point Cloud, Global Mapper, LAStools, Overture Maps, Cloud-Optimized GeoTIFF, Folium, FME

Cloud & DevOps: AWS Lambda, S3, SageMaker, ECR / ECS, AWS CDK, Docker, IAM, VPC / Network Security, CI/CD

Field: GNSS / GPS Survey, UAV Survey (FAA Part 107), Differential GNSS, Total Station and Leveling Survey, Geodatabase Design

EXPERIENCE

Data Scientist (Geospatial & AI) — Apr 2025 – Aug 2026

Florida Power & Light - Contract, West Palm Beach, FL

- Defined ML scope and technical specs as data scientist/supporting product owner for a multi-million-dollar AI vendor engagement; authored RFP technical requirements and served as technical SME through development and benchmarking.
- Independently built and deployed an overhead switch-state (open/closed) classification model, training it locally then migrating to AWS (ECR, SageMaker) for production inference.
- Built an API-based metadata pipeline with spatial (10m/30ft) and temporal (30-day) joins matching 350K+ condition reports to an 8M-image catalog, enriching ~2M images with condition metadata.
- Acted as product owner for a dataset-creation plugin built with the enterprise IT team, defining filters, UI, and delivery logistics, and used it to deliver 300+ curated imagery datasets to vendors.
- Built and deployed GIS applications (ArcGIS Enterprise, Experience Builder, Dashboards, Utility Network) for infrastructure monitoring and disaster response, including spatial web portals used in real time during Hurricane Milton for structural impacts, outages, and crew deployment.

Associate Geospatial Analyst — Sep 2024 – Mar 2025

Florida Power & Light - Contract, Jupiter, FL

- Led QA/QC of LiDAR-derived power grid data for infrastructure resilience work, including pole hardening.
- Managed PostgreSQL/PostGIS databases, optimized Cloud-Optimized GeoTIFFs, and automated QA workflows in FME, ArcPy, and Model Builder to streamline data storage and validation.

Research Assistant — Jan 2023 – Aug 2024

Florida Atlantic University, Boca Raton, FL

- Conducted FEMA hydrologic and hydraulic (H&H) flood risk assessments for 3 Florida communities using Python automation, part of FAU's $1.7M FEMA-funded CWR3 Watershed Master Planning Initiative to build a statewide planning template.
- Built a Python geocoding tool processing 3 million addresses, replacing a paid service; cut turnaround from 3 months to 1 week and saved $20K in licensing.

Geomatics Engineer Dec 2020 – Dec 2022

Freelance Consultant, Kathmandu, Nepal

- Led survey teams for municipal road, control, and construction surveys, including GNSS landmark and UAV topographic/transmission-line surveys, and drone photogrammetry for orthomosaic mapping.
- Designed municipal geodatabases (land use, soil, cadastral, resource maps) with topology and attribute rules.

EDUCATION

MSc, Geosciences — GIS Concentration — Florida Atlantic University · 2023–2024

BSc, Geomatics Engineering — Tribhuvan University, Nepal · 2016–2021

CERTIFICATIONS & TRAINING

- The Rise of the AI Data Engineer Bootcamp (2026)
- FAA Part 107 — Remote Pilot Certificate
- Machine Learning Specialization — DeepLearning.AI
- Deep Learning Specialization — DeepLearning.AI
- Remote Sensing Image Acquisition & Analysis — UNSW Canberra
- GIS Mapping & Spatial Analysis — U. Toronto
- Data Science with R — Johns Hopkins
- GIS Specialization — UC Davis
- Introduction to Databases — Meta
- **NASA ARSET Certifications (12):** Applied remote sensing training (2022-2024) in climate, disaster, and environmental risk modeling.
- **Esri Training Certifications (8):** GIS fundamentals, cartography, spatial data science, and imagery analysis.

PUBLICATIONS

Dhungana, B.; Liu, W. *Urban–Rural Exposure to Flood Hazard and Social Vulnerability in the Conterminous United States*. ISPRS Int. J. Geo-Inf. 2024, 13, 339.

Mandal, A.K.; Thapa, A.K.; Thapa Chhetri, M.; Dhungana, B.; Bloetscher, F.; Yong, Y.; Su, H. *PyWMP: A Unified Python Framework for Multi-Fidelity 1D, 2D, and Hybrid Event-Based Flood Modeling*. Manuscript in review, DOI pending; ongoing contributor to the open-source PyWMP package.

PROJECTS

SurgeExposure: A cloud-native pipeline overlaying NOAA storm-surge and flood-inundation data against Overture Maps building footprints, scoring per-asset flood exposure across 8 coastal regions — with a map and API. *(Python, DuckDB Spatial, FastAPI, PostGIS, AWS)* Demo | Repo

fflood-nep: A reproducible flood-response pipeline for Nepal's August 2026 Rasuwa/Bhote Koshi flash flood — runs Sentinel-1 SAR change detection for flood extent, scores building/road/facility exposure by municipality, and tracks a separate InSAR pipeline for landslide/avalanche source detection. Built and shipped inside the event's first week. *(Python, Sentinel-1 SAR, STAC, GDAL, InSAR)* Demo | Repo

wildfire-copilot: A LangGraph agent for utility wildfire risk — combines a trained XGBoost susceptibility model with live NASA/NWS hazard feeds and RAG over real wildfire-mitigation filings to answer plain-English risk questions and prioritize inspections. *(LangGraph, XGBoost, RAG, FastAPI)* Demo | Repo

geomlint: A CI-first data-quality linter for vector geospatial files — catches invalid geometry, CRS confusion, and topology errors that mainstream data-observability tools skip. No GDAL required; pip install and it runs anywhere pip does. *(Python, Shapely, PyProj, PyPI, CLI)* Docs | Repo

Road-Centerline: A published Python package that extracts road centerlines from polygon geometries — worldwide, in any input CRS, in any format GeoPandas can read. CLI and Python API, computed via medial-axis skeletonization. *(Python, GeoPandas, pygeoops, PyProj, PyPI)* Repo